What Survives of the Split–Forest Idea

A Companion Note to the Repaired Decidability Proof for $\mathcal{ALCI}_{\mathrm{RCC5}}$

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Abstract

This note explains which parts of the original split–forest proof idea remain intact after the proof is repaired using witness thinning, generated cover presentations, and finite regular certificates. The main conclusion is that the split–forest idea is not discarded. It is reinterpreted. The tree is not the Hasse diagram of an arbitrary semantic PP/PPI order, and it is not required to have an ambient semantic root. Instead, it is a generated witness-cover presentation: selected containment edges form a tree-like skeleton; split copies remove multi-parent obstructions; sibling interactions are stored in support-closed mosaics; typed EQ is restored by an explicit congruence; and the finite object is a regular certificate whose unfolding may be infinite.

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1 Purpose of this note

The repaired proof changes the formal reading of the original split–forest construction. This can make it look as if the original idea has been replaced. That is not the right interpretation. Most of the original architecture survives, but several phrases must be read in the generated/tableau sense rather than in the semantic Hasse-diagram sense.

The core correction is:

A cover edge is a generated witness-cover edge, not a semantic Hasse cover.

Once this distinction is made, the original proof strategy remains recognizable. The repaired proof still uses a containment backbone, split copies, sibling interfaces, DR/PO frontier simplification, typed EQ quotienting, and finite quotient enumeration. What changes is the status of the finite object: it is a finite regular certificate, not a finite semantic model and not a finite rooted Hasse diagram.

2 The original intuition

The original split–forest idea can be summarized as follows.

1. The difficult PP/PPI part of RCC55 should be represented vertically, by ancestor and descendant relations in a tree or forest.
2. If a semantic element appears to need several containment parents, it should be split into several copies so that the presentation remains tree-shaped.
3. Non-vertical interactions, especially interactions between sibling subtrees, should be recorded by finite interface data.
4. Equality should first be handled weakly, through synchronized copies, and then restored by a strong typed quotient.
5. Since only finitely many local closure types and finite interface patterns are relevant, satisfiability should reduce to the existence of a finite quotient or certificate.

All five ideas still survive. The repaired proof changes the interpretation of items 1 and 5, and strengthens items 3 and 4.

3 The key reinterpretation: generated witness covers

The repaired proof does not begin by taking an arbitrary semantic RCC55 model and forming the Hasse diagram of its PP order. That would be the wrong object. An arbitrary semantic model may contain irrelevant elements between selected witnesses, and those elements need not matter for the satisfaction of the input concept.

Instead, the proof first passes to a witness-generated abstract model.

Definition 1 (Witness-generated model, informal)**.** Let C_0 be the input concept and let $\mathrm{Cl}(C_0)$ be its closure under subformulas and negation normal form complement. A pointed model $(\mathcal{I}, a_0)$ is witness-generated for C_0 if every element of its domain is reachable from a_0 by repeatedly choosing witnesses for existential formulas

$$\exists R.D \in \mathrm{Cl}(C_0), \qquad R \in \{\mathsf{PP}, \mathsf{PPI}, \mathsf{PO}, \mathsf{DR}, \mathsf{EQ}\}.$$

For each active existential requirement at a generated node, the construction retains a selected witness.

The generated edge relation is not the semantic PP or PPI relation itself. It is a selected presentation relation. For example,

$$x \xrightarrow{\mathsf{PP}}_{\mathsf{gen}} y$$

means that y is a selected generated witness for a PP-existential at x. Semantically this implies $x\mathsf{PP}y$, but it does not assert that y is an immediate semantic Hasse successor of x.

This removes the earlier density issue. Density of an arbitrary ambient semantic order is irrelevant, because the proof works with a sparse generated submodel containing the witnesses needed by the logic.

4 Why the two preliminary steps recover the intended model class

It is helpful to separate the intended generated-model intuition from the fully general semantic formulation of satisfiability. The repaired proof starts from the standard statement

$$C_0 \text{ is satisfiable in some abstract RCC55 model,}$$

and then uses two preliminary conservativity steps to reach the sparse model class originally intended by the split–forest construction.

4.1 First step: witness thinning

The first step is the witness-generated submodel lemma. Given any satisfying pointed model $(\mathcal{I}, a_0)$, the construction retains only the elements that are needed for the closure formulas of C_0. Starting from a_0, whenever a generated element a satisfies an existential requirement

$$\exists R.D \in \mathrm{Cl}(C_0),$$

one selected witness b with aRb and $b \models D$ is kept. Iterating this process gives a generated subdomain.

This step is not primarily a geometric or topological claim. It is a modal/model-theoretic thinning argument. Existential formulas remain true because their selected witnesses are retained. Universal formulas remain true because restricting the domain removes possible successors; it cannot introduce new counterexamples to a universal restriction. With closure under negation-normal-form complement, truth of all closure formulas is preserved.

Thus, if C_0 is satisfiable at all, then it is satisfiable in a model where every element is present for a logical reason: it is either the evaluation point or a selected witness for some existential closure formula. This is exactly the sparse, generated Kripke-style model class that the original split–forest idea was aiming at.

The earlier discussion of dense or non-Hasse PP/PPI orders should be understood only as a warning about arbitrary starting models. Such models may contain irrelevant intermediate elements. Witness thinning makes those elements irrelevant. After thinning, the proof works with selected witnesses, not with every element of the original semantic order.

4.2 Second step: generated cover presentation

The second step is to turn the thinned witness-generated model into a containment-oriented generated cover presentation. The proof introduces an auxiliary edge relation such as

$$x \xrightarrow{\mathsf{PP}}_{\mathsf{gen}} y$$

meaning that y is a selected generated witness for a PP-requirement at x, or dually a selected containment witness in the split presentation. This generated edge is immediate in the presentation. It is not required to be an immediate cover in the original model before thinning.

The direction of construction is therefore the reverse of a semantic Hasse-diagram construction. A Hasse diagram starts with a complete transitive order and extracts its immediate-cover relation. The repaired split–forest proof starts with selected generated cover edges and interprets vertical semantic containment by transitive closure along branches:

$$u\mathsf{PP}v \quad \text{when} \quad v \text{ is a strict generated superpart ancestor of } u.$$

In the final reconstructed split model, the vertical edges are Hasse-like for the generated containment presentation. But they are not obtained by taking the Hasse diagram of the arbitrary original model. They are chosen presentation edges produced after witness thinning and splitting.

4.3 Relation to the original intention

With these two preliminary steps in place, the repaired proof reaches essentially the model class originally envisioned:

sparse abstract Kripke structures generated by logical witness requirements.

The proof only begins from arbitrary abstract models because the usual satisfiability theorem is stated that way. The thinning lemma and generated-cover presentation show that this broader starting point is conservative: no satisfiable concept is lost by restricting attention to the intended generated structures.

So the corrected reading is not that the original split–forest proof must handle arbitrary dense semantic orders directly. Rather:

$$\boxed{\text{arbitrary model} \implies \text{witness-generated submodel} \implies \text{generated split–forest presentation.}}$$

Once this reduction is made, more of the original proof idea survives. The containment backbone can be treated as a generated cover tree; split copies handle multiple generated containment placements; sibling interfaces handle non-vertical DR/PO and core interactions; and the finite quotient is a regular certificate for the generated presentation.

If one takes witness-generated abstract Kripke structures as the semantics from the outset, then the first two steps are not additional construction steps. They become the formal justification that the intended semantics is conservative over the ordinary existential satisfiability statement. In either presentation, the proof works over the same sparse generated split–forest objects.

5 The containment-oriented split tree

The repaired proof keeps the original containment-oriented reading. There is one vertical containment orientation. For definiteness:

$$\text{parent} = \text{proper superpart}, \qquad \text{child} = \text{proper part}.$$

Thus, along a vertical branch,

$$\text{child } \mathsf{PP} \text{ parent}, \qquad \text{parent } \mathsf{PPI} \text{ child}.$$

More generally, semantic PP/PPI between comparable tree occurrences is obtained by transitive closure along the branch.

This keeps the original separation:

$$\boxed{\text{vertical branch structure handles } \mathsf{PP}/\mathsf{PPI}.}$$

The repaired proof only removes two overly strong readings:

$$\text{tree edge} \neq \text{semantic Hasse cover},$$

and

$$\text{tree root} \neq \text{ambient top element required by the semantics}.$$

The split tree is rooted as a generated presentation when possible, but the relevant containment trunk may be infinite upward, infinite downward, or bi-infinite around the evaluation point. This is why the finite proof object is regular rather than literally finite.

6 What survived

6.1 The tree-shaped containment backbone survives

The first surviving idea is the most important one. The repaired proof still represents PP and PPI vertically.

In the repaired proof, a vertical branch

$$\cdots \prec u_2 \prec u_1 \prec u_0$$

represents a chain of proper-part/proper-superpart relations. If u_{i+1} is the parent of u_i, then

$$u_i \mathsf{PP} u_{i+1} \qquad \text{and} \qquad u_{i+1} \mathsf{PPI} u_i.$$

By transitivity, if $i < j$ then

$$u_i \mathsf{PP} u_j.$$

What changes is that this branch is a generated presentation. It need not be the Hasse diagram of a pre-existing semantic partial order. The branch edges are selected witness-cover links. They are sparse by construction.

6.2 Split copies survive

Split copies also survive essentially unchanged.

The reason is structural. A tree presentation cannot literally give one node several parents. But generated containment requirements can point to several selected superparts, and different sibling contexts may require the same semantic element to be seen from different containment positions. The solution remains the original one: make several copies in the weak split presentation.

These copies are not automatically semantically equal. They are presentation copies. If the proof intends them to represent one strong-EQ element, this must later be justified by the typed EQ congruence.

Thus the repaired proof keeps the split-copy principle:

$$\boxed{\text{Split first; quotient by typed } \mathsf{EQ} \text{ only when congruence is certified.}}$$

6.3 Sibling interfaces survive, strengthened as mosaics

The sibling-interface idea survives, but the raw pair-table version is too weak. The repaired proof uses support-closed mosaics.

The reason is witness coherence. Suppose a type p can have a child of type p' in one occurrence, and a relation R to a type q in another occurrence. A raw pair table may record both facts and then incorrectly combine them as if one occurrence witnessed both. This is the witness-conflation problem.

A support-closed mosaic records a finite joint pattern, not merely independent pair facts. It stores enough information to certify that the relevant pair and triple labels can occur together.

Thus the original interface idea survives in this stronger form:

$$\boxed{\text{sibling interactions are finite, support-closed joint mosaics.}}$$

6.4 The DR/PO sibling simplification survives with qualifications

The original proof's useful simplification was that ordinary non-core sibling frontier nodes do not need arbitrary $\mathsf{PP}/\mathsf{PPI}/\mathsf{EQ}$ cross-links. After containment and core/equality interactions have been accounted for, the remaining open sibling relation can be taken from

$$\{\mathsf{DR}, \mathsf{PO}\}.$$

This survives, but only in the qualified form:

$$\boxed{\text{non-core sibling frontier edges are } \mathsf{DR}/\mathsf{PO} \text{ after support closure and core handling.}}$$

It should not be stated as "all sibling interactions are automatically DR/PO." Core nodes, equality mates, and composition-induced containment constraints still need explicit bookkeeping.

6.5 Typed EQ quotienting survives

The repaired proof keeps the weak-to-strong equality strategy. In the weak split presentation, multiple copies may be linked as intended EQ-mates. Strong equality is restored only after quotienting by an explicit congruence relation $\equiv_{\mathsf{EQ}}$.

The congruence must guarantee at least the following.

1. If $x \equiv_{\mathsf{EQ}} y$, then x and y have the same closure type.

2. They have matching witness obligations and universal obligations.
3. They have the same externally visible RCC55 relations to all other equivalence classes.
4. The quotient does not identify points related by strict PP or strict PPI.

So the original typed-EQ idea survives, but it must be explicit. In particular, blocking repetition is not EQ.

6.6 The finite quotient survives as a regular certificate

The original finite quotient idea survives, but its interpretation changes.

The finite object is not itself a semantic model. It is a finite regular certificate whose unfolding may be infinite. A cyclic edge in the certificate means:

$$\text{repeat this profile with fresh occurrences,}$$

not

$$\text{identify the repeated occurrence by } \mathsf{EQ}.$$

For example, the satisfiable concept

$$\exists \mathsf{PP}.\top \sqcap \forall \mathsf{PP}.\exists \mathsf{PP}.\top$$

may be represented by a one-profile cycle. Its unfolding is

$$u_0 \mathsf{PP} u_1 \mathsf{PP} u_2 \mathsf{PP} u_3 \mathsf{PP} \cdots ,$$

with all u_i distinct. The cycle is a blocking cycle in the finite certificate, not a semantic PP cycle.

Thus the correct surviving statement is:

$$\boxed{\text{satisfiability is equivalent to a valid finite regular split–forest certificate.}}$$

7 What was replaced or discarded

The following original readings should not be used in the repaired proof.

7.1 No semantic Hasse diagram

Tree edges should not be defined as immediate covers of the semantic PP/PPI order. The proof constructs selected witness-cover edges. These edges generate the presentation, and semantic PP/PPI is obtained by closure along the branch.

7.2 No ambient semantic root

The proof should not require every relevant containment branch to terminate at a semantic top or bottom region. A concept can force an infinite chain of proper superparts or proper parts. The repaired proof therefore allows regular trunks and blocking cycles.

7.3 No well-foundedness assumption

The repaired proof uses acyclicity and transitivity of strict PP/PPI, not well-foundedness of either order. Infinite generated branches are handled by regularity and eventuality checks.

7.4 No raw pair-table completeness

Raw pair tables do not preserve joint realizability. They are replaced by support-closed mosaics with explicit restriction and triple-composition coherence.

7.5 No blocking-as-equality

Blocking equivalence and semantic equality must remain separate:

$$\equiv_{\mathsf{blk}} \neq \equiv_{\mathsf{EQ}} .$$

A blocked cycle unfolds to fresh semantic occurrences. Only the typed EQ congruence may be quotiented.

8 How the repaired proof uses the surviving ideas

The repaired proof can be read as the following pipeline.

Step 1: Closure and types. Compute the finite closure $\mathrm{Cl}(C_0)$, including NNF complements. Define Hintikka types over this closure.

Step 2: Witness-generated submodel. Show that if C_0 is satisfiable, then it is satisfiable in a witness-generated abstract RCC55 model. All existential closure formulas have selected witnesses in the generated submodel. This step removes irrelevant ambient material.

Step 3: Containment split presentation. Present the generated PP/PPI witnesses by a containment-oriented split tree. Parent means proper superpart; child means proper part. Multiple selected superparts are handled by split copies.

Step 4: Interface mosaics. Represent non-vertical DR/PO and equality/core interactions by finite support-closed mosaics. These mosaics prevent pairwise witness conflation and enforce the RCC55 composition table locally.

Step 5: Regular quotient. Because there are finitely many closure types, finite context descriptors, finite equality colors, and finite mosaic shapes, a valid infinite split presentation has a finite regular certificate. Cycles in this certificate are blocking cycles.

Step 6: Eventuality checks. Transitive PP/PPI existential obligations are checked by reachability in the finite certificate. A cyclic trunk is valid only if every eventuality occurring on the cycle is realized somewhere later in the same cycle, or along a reachable continuation.

Step 7: Typed EQ quotient. Unfold the finite certificate into a weak split model. Then quotient only by the certified typed-EQ congruence. The quotient is a strong-EQ RCC55 model satisfying C_0.

9 Comparison table

Original component	Repaired status
Tree-shaped containment backbone	Survives as a generated witness-cover tree, not as a semantic Hasse diagram.
Split copies	Survive; they are needed for elements with several selected containment parents or several interface occurrences.
Sibling interfaces	Survive; they must be support-closed mosaics rather than raw pair tables.
DR/PO sibling simplification	Survives for non-core sibling frontier nodes after containment, core, equality, and support closure are accounted for.
Typed EQ quotient	Survives; it must be an explicit congruence, separate from blocking.
Finite quotient	Survives as a finite regular certificate, not as a finite semantic model.
Ambient semantic root	Removed. Regular trunks replace root termination.
Semantic Hasse covers	Removed. Generated witness-cover edges replace them.
Well-foundedness	Not assumed. Infinite branches are handled by regularity and eventuality checks.
Raw pair tables	Replaced by support-closed mosaics.
Blocking as EQ	Rejected. Blocking repeats profiles but unfolds to fresh occurrences.

10 Recommended wording

A concise way to state the repaired split–forest theorem is:

> Every satisfiable $\mathcal{ALCI}_{\mathrm{RCC5}}$ concept has a witness-generated containment-oriented regular split–forest presentation. In this presentation, vertical generated witness-cover edges represent selected PP/PPI containment links; semantic PP/PPI is obtained by transitive closure along branches; sibling and frontier interactions are recorded by finite support-closed mosaics; weak equality copies are quotiented only by an explicit typed congruence; and cyclic blocking in the finite certificate unfolds to fresh semantic occurrences.

This wording keeps the original split–forest idea while avoiding the invalid readings that caused difficulty.

11 Summary

With the witness-generated submodel lemma in place, the proof no longer needs to worry about dense ambient PP/PPI orders or semantic Hasse diagrams. This means more of the original split–forest idea survives than might first appear.

The repaired proof is still a split–forest proof. Its backbone is still containment-oriented. It still uses splitting, sibling interfaces, DR/PO frontier simplification, typed EQ, and finite quotient enumeration. The essential repair is that the finite object is now understood as a generated regular certificate with support-closed interfaces and explicit equality congruence.

In one sentence:

$$\boxed{\text{The original split–forest idea survives as a generated regular split–forest certificate theorem.}}$$